Cardiff Metropolitan University

Topic

Leveraging Analytics and GIS for Hotel & Tourism Sector Development in Sri Lanka

A Data-Driven Approach for Informed Decision Making

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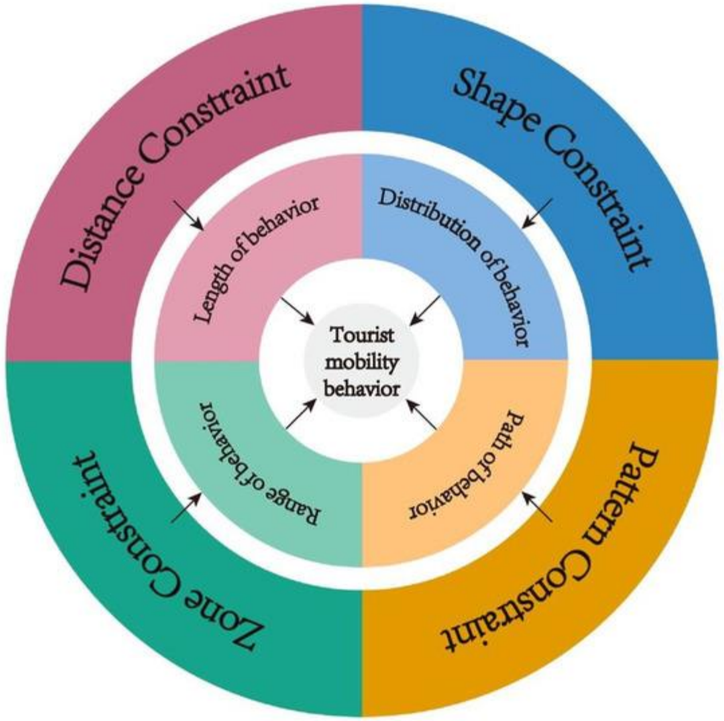
Introduction to the Topic

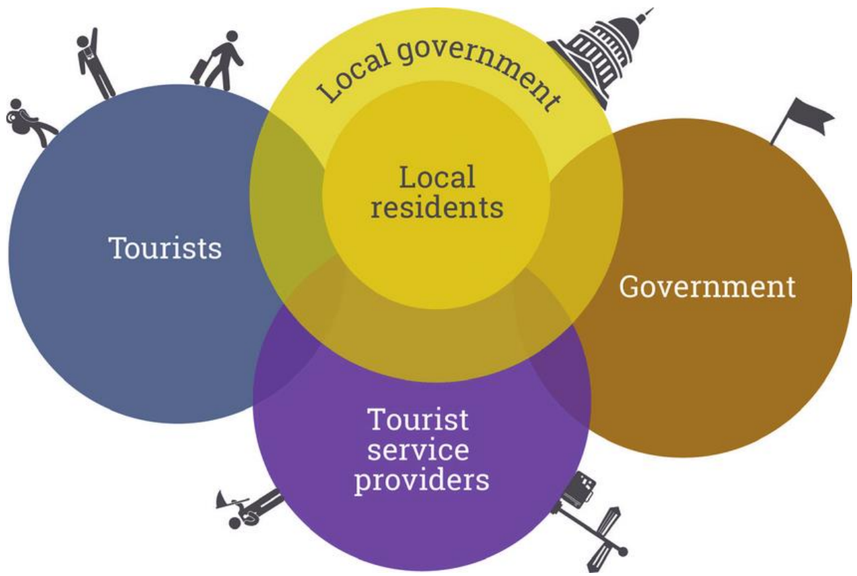
- **Utilizing Analytics and Business Intelligence (ABI) for the Tourism sector in Sri Lanka**
- **Merging Statistical applications with Geographic Information Systems (GIS)**
- **Enhancing decision-making for government and private sector institutions**
- **Transforming raw data into actionable intelligence to solve industry problems**

Importance of the Topic (Global & Local Trends)

- **Global Trend:** Shift towards 'Smart Tourism' using location-based services
- **Local Context:** Tourism is a primary foreign income generator for Sri Lanka
- **Trend Integration:** Need for high-level administrative officials to eliminate ambiguity in planning
- **Sustainability:** Using BI to ensure long-term development across different parts of the island.

Issues in the Current Sector

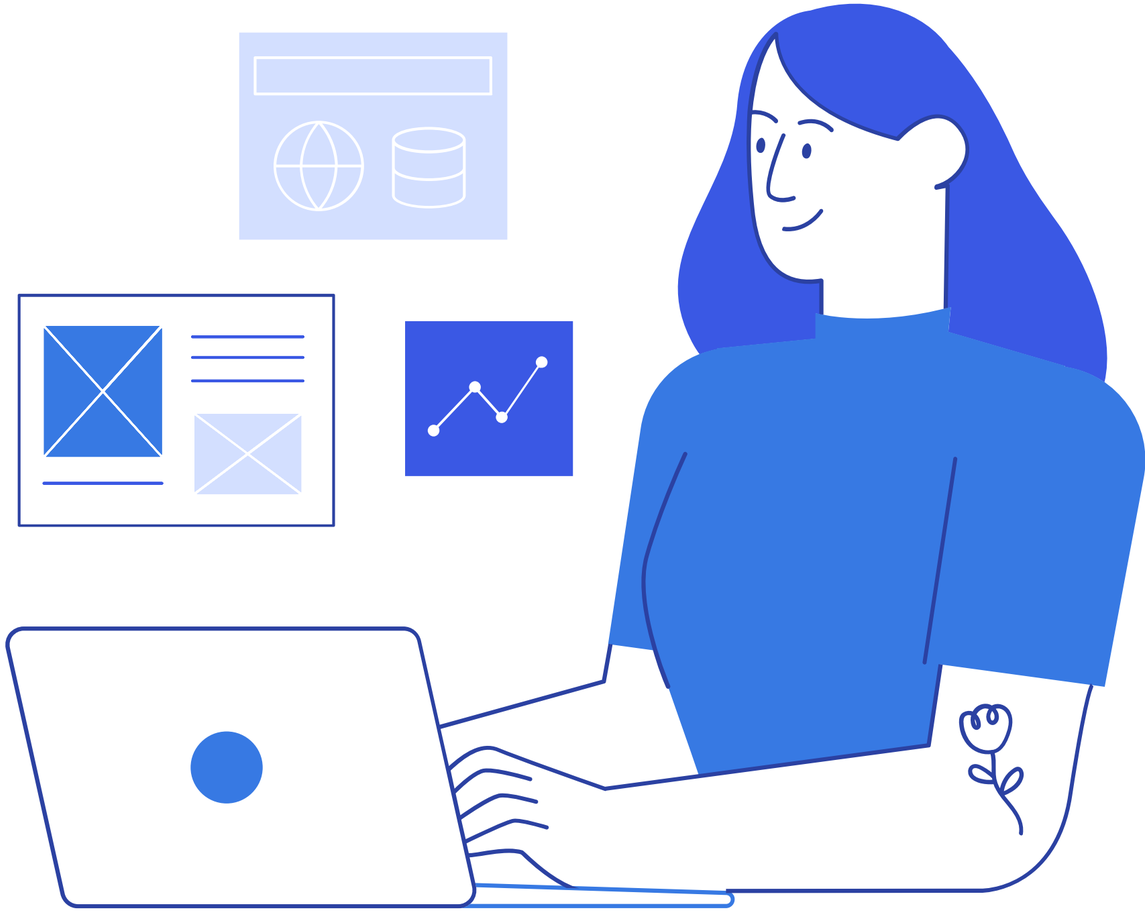




- **Information Gaps:** Lack of precise data on tourist spatial behavior
- **Uncertainty:** High levels of ambiguity in traditional decision-making processes.
- **Resource Allocation:** Inefficient distribution of infrastructure based on outdated models
- **Income Leakage:** Missing opportunities to capture maximum foreign revenue

Opportunities for BI Intervention

- **Data-Driven Growth:** Escalating development through identified new opportunities.
- **Targeted Marketing:** Using analytics to identify high-spending foreign demographics
- **Strategic Positioning:** Placing tourism hubs where they generate maximum efficiency.
- **Policy Support:** Aiding the Ministry of Foreign Affairs in income-generating strategies



Literature Review - Statistical BI

01

Predictive Analytics:

Research shows statistical models can forecast arrival trends.

02

Quantitative Methods:

Using historical data to identify growth patterns

03

Software Landscape:

Application of licensed and open-source statistical software

04

BI Maturity:

How data analysis leads to better 'Informed Decision Making'

Literature Review - GIS in Tourism

01

Spatial Intelligence:

Understanding tourism through 'Geospatial' lenses

02

Visualizing Trends:

Using maps to identify tourism clusters and hotspots

03

Historical Context:

Reviewing how GIS has been used globally for destination management

04

Tool Evolution:

The shift from simple maps to complex spatial data models

Proposed Methodology Overview

- **Hybrid Approach:** Combining Geospatial and Non-Geospatial data models
- **Analytical Pipeline:** Data Extraction -> Processing -> Intelligence Generation
- **Applicability:** Ensuring the methodology fits the local Sri Lankan context
- **Outcome:** A proposal for solving problems and improving the present status

Data Models - Non-Geospatial Data

Demographic Data:

Age, nationality, and spending habits of tourists.

Temporal Data:

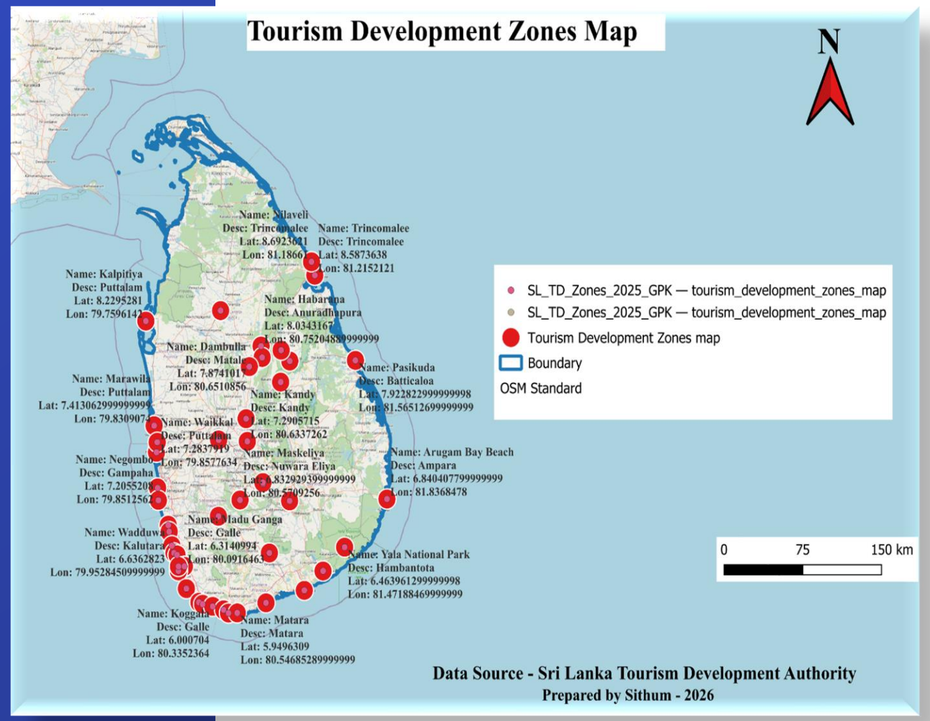
Arrival times and duration of stay

Economic Indicators:

Exchange rates and hotel occupancy percentages

Feedback Data:

Qualitative reviews and satisfaction scores



Data Models - Geospatial Data

Vector Data Models:

Locations of hotels (points) and transport networks (lines)

Raster Data Models:

Land use patterns and environmental satellite imagery

Coordinate Systems:

Using GPS data to track popular tourist routes

Spatial Layers:

Overlaying tourism zones with infrastructure maps

09

Statistical Tools & Techniques

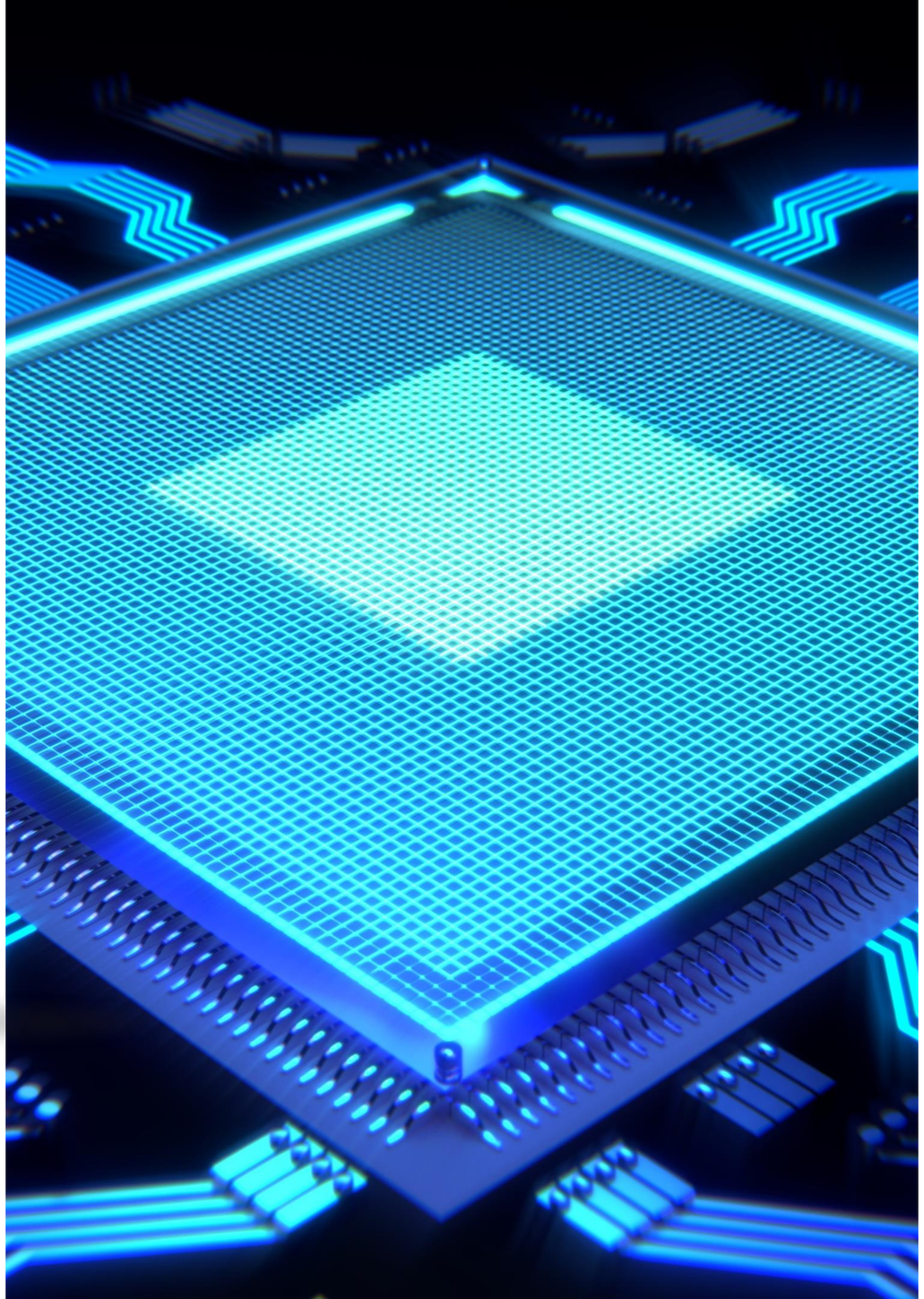
- **Descriptive Statistics:** Summarizing tourist arrivals and income metrics

SPSS DATA ANALYSIS

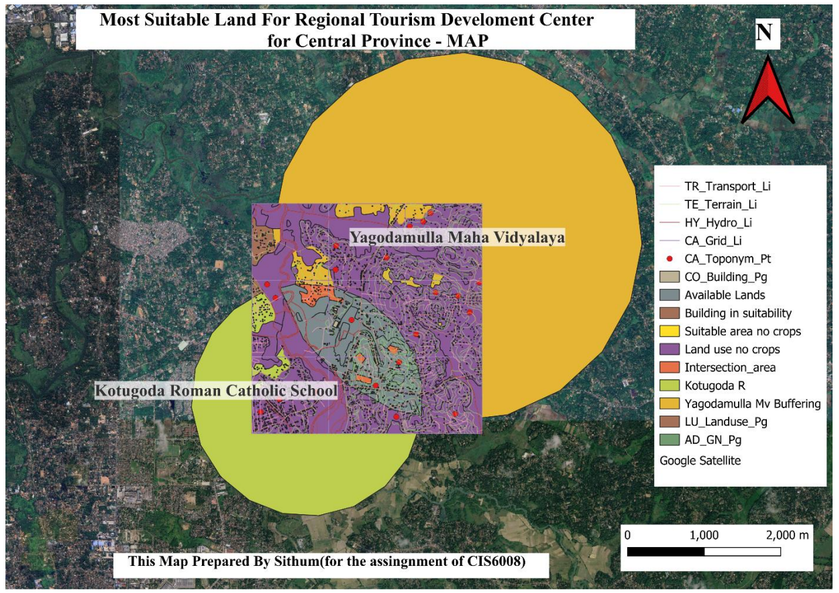
- **Regression Analysis:** Determining factors that influence foreign income growth.

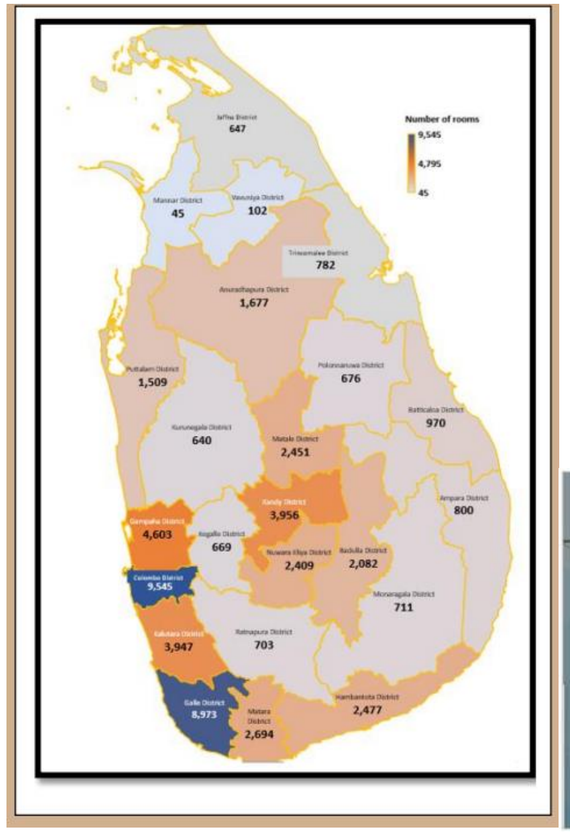
- **Forecasting:** Predicting future trends for the Ministry of Tourism

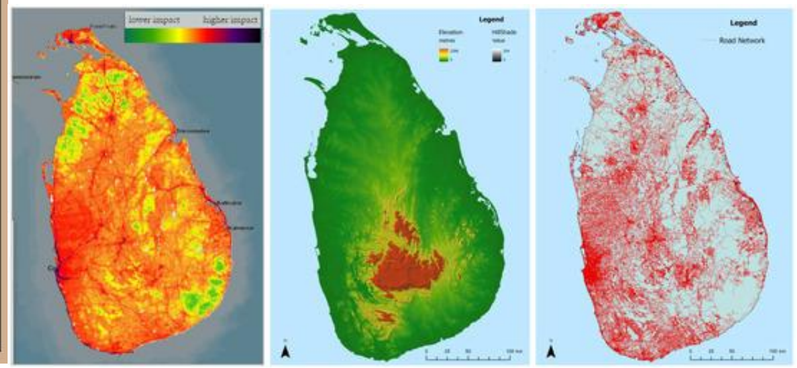
- **Tools:** Utilization of software like SPSS, R, or Excel for analysis



Geospatial Tools & Techniques







- **Buffer Analysis:** Evaluating the proximity of hotels to attractions.
- **Heat Maps:** Identifying high-density areas of tourist activity
- **Network Analysis:** Optimizing travel routes for foreign transport efficiency
- **GIS Software:** Implementation using tools like ArcGIS or QGIS

Generating Business Intelligence (BI)

01 **Synthesis:**
Merging maps with statistics to create 'Business Intelligence'

02 **Decision Support:**
Providing clear visuals for Ministry officials

03 **Strategic Advantage:**
Gaining an edge over regional tourism competitors

04 **Reliability:**
Reducing discrimination and bias in government planning

Ethical Considerations (EDGE Skills)

01 **Data Ethics:**
Adhering to formal ethical practices during data extraction

02 **Privacy:**
Ensuring tourist data is processed and disseminated legally

03 **University Guidelines:**
Following established academic integrity procedures

04 **Integrity:**
Maintaining transparency in reporting statistical findings

Entrepreneurial & Global Perspectives

- **Global Benchmarking:** Localizing international BI findings for Sri Lanka
- **New Ventures:** Motivating students/firms to initiate BI consultancy services
- **Economic Escalation:** Creating solutions for corporate and individual customers
- **Innovation:** Using BI to identify niche tourism markets (e.g., Eco-tourism)

Feasibility of the Proposal

Technical Feasibility:

Availability of open-source GIS and statistical tools

01

02

Operational Ease:

Officials can use these tools to generate maximum benefits

Scalability:

Proposal can be expanded to other sectors beyond tourism

03

04

Cost-Benefit:

High ROI by eliminating wasteful spending through BI

Implementation Strategy

01

Collaboration:

Working with Ministry of Foreign Affairs & Tourism

02

Data Centralization:

Creating a unified database for geospatial and non-geospatial data

03

Training:

Capacity building for administrative officials

04

Monitoring:

Continuous data updates to reflect real-time trends

Recommendations

01

Adopt GIS:

Fully integrate spatial mapping into the national tourism master plan

02

Invest in BI:

Allocate budget for advanced analytical technologies

03

Public-Private Data Sharing:

Improve data accuracy for better intelligence

04

Focus on Foreign Income:

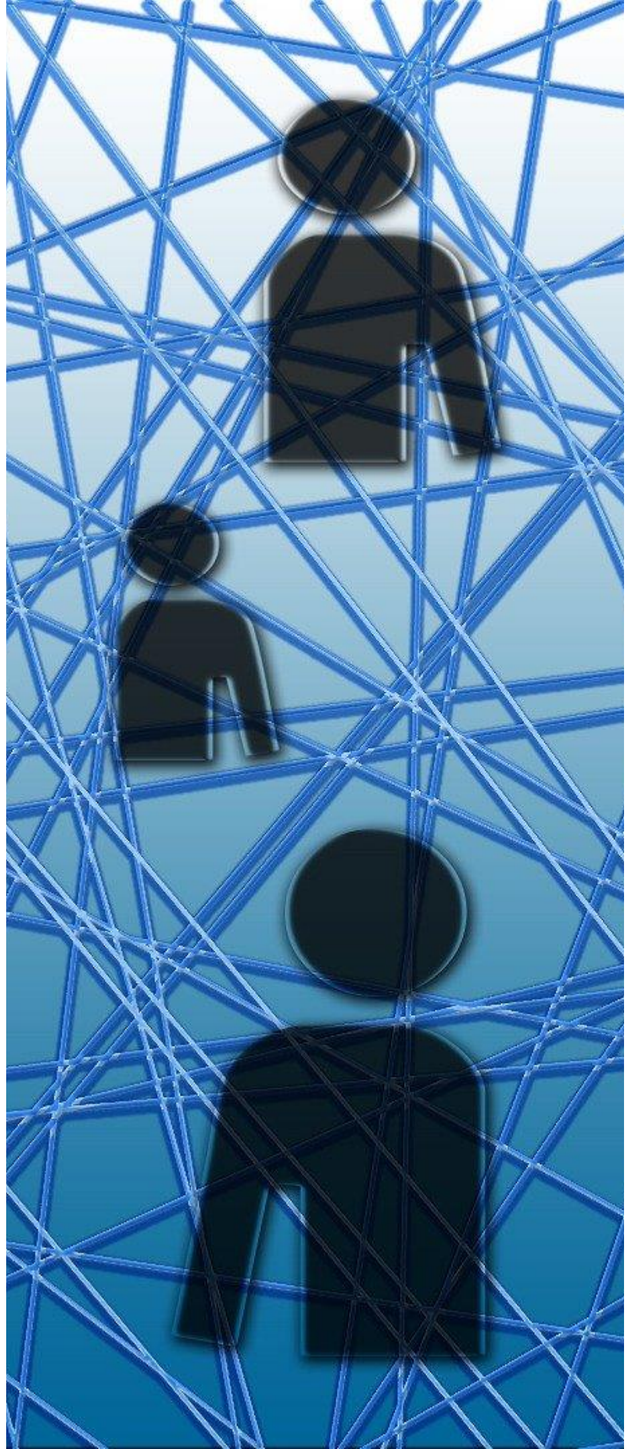
Prioritize strategies that maximize foreign currency inflow

Conclusion

- **Summary:** Statistical and GIS tools are vital for informed decision-making
- **Final Impact:** These technologies eliminate uncertainty and drive sector development.
- **Future Outlook:** ABI will be the cornerstone of Sri Lanka's tourism success.

Q &A

Thank You!



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